

REFERENCE: SIMPLY-TYPED λ -CALCULUS

Samantha Frohlich

Figure 1: Statics of the constants-and-functions fragment of the simply-typed λ -calculus

$$\begin{array}{c}
\text{VAR} \\
\hline
\Gamma, x : \sigma \vdash x : \sigma
\end{array}
\quad
\begin{array}{c}
\text{NUM} \\
n \in \mathbb{N} \\
\hline
\Gamma \vdash \text{num}[n] : \text{Num}
\end{array}
\quad
\begin{array}{c}
\text{PLUS} \\
\Gamma \vdash e_1 : \text{Num} \quad \Gamma \vdash e_2 : \text{Num} \\
\hline
\Gamma \vdash \text{plus}(e_1; e_2) : \text{Num}
\end{array}$$

$$\begin{array}{c}
\text{TIMES} \\
\Gamma \vdash e_1 : \text{Num} \quad \Gamma \vdash e_2 : \text{Num} \\
\hline
\Gamma \vdash \text{times}(e_1; e_2) : \text{Num}
\end{array}
\quad
\begin{array}{c}
\text{LET} \\
\Gamma \vdash e_1 : \sigma_1 \quad \Gamma, x : \sigma_1 \vdash e_2 : \sigma_2 \\
\hline
\Gamma \vdash \text{let}(e_1; x. e_2) : \sigma_2
\end{array}$$

$$\begin{array}{c}
\text{LAM} \\
\Gamma, x : \sigma \vdash e : \tau \\
\hline
\Gamma \vdash \lambda x : \sigma. e : \sigma \rightarrow \tau
\end{array}
\quad
\begin{array}{c}
\text{APP} \\
\Gamma \vdash e_1 : \sigma \rightarrow \tau \quad \Gamma \vdash e_2 : \sigma \\
\hline
\Gamma \vdash e_1(e_2) : \tau
\end{array}$$

Figure 2: Dynamics of the constants-and-functions fragment of the simply-typed λ -calculus

$$\begin{array}{c}
\text{VAL-LAM} \\
\hline
\lambda x : \tau. e \text{ val}
\end{array}
\quad
\begin{array}{c}
\text{VAL-NUM} \\
n \in \mathbb{N} \\
\hline
\text{num}[n] \text{ val}
\end{array}
\quad
\begin{array}{c}
\text{D-PLUS} \\
n_1 + n_2 = n \\
\hline
\text{plus}(\text{num}[n_1]; \text{num}[n_2]) \mapsto \text{num}[n]
\end{array}$$

$$\begin{array}{c}
\text{D-PLUS-1} \\
e_1 \mapsto e'_1 \\
\hline
\text{plus}(e_1; e_2) \mapsto \text{plus}(e'_1; e_2)
\end{array}
\quad
\begin{array}{c}
\text{D-PLUS-2} \\
e_1 \text{ val} \quad e_2 \mapsto e'_2 \\
\hline
\text{plus}(e_1; e_2) \mapsto \text{plus}(e_1; e'_2)
\end{array}$$

$$\begin{array}{c}
\text{D-LET} \\
\hline
\text{let}(e_1; x. e_2) \mapsto e_2[e_1/x]
\end{array}
\quad
\begin{array}{c}
\text{D-APP-1} \\
e_1 \mapsto e'_1 \\
\hline
e_1(e_2) \mapsto e'_1(e_2)
\end{array}
\quad
\begin{array}{c}
\text{D-BETA} \\
\hline
(\lambda x : \tau. e_1)(e_2) \mapsto e_1[e_2/x]
\end{array}$$